

Incident Report: SSH Brute Force Attack Against Ubuntu Target

Report Metadata

- Incident ID: INC-2026-001
- Date of Activity: October 3, 2026
- Report Author: Trevor
- Classification: Lab Exercise (Home SIEM Environment)
- Severity: High (Successful Credential Compromise)

1. Executive Summary

On October 3, 2026, a simulated SSH brute-force attack was executed against the monitored Ubuntu target host (10.0.0.92) in a home-lab SIEM environment. The attack successfully identified valid credentials for the user account yboxuser after a series of failed authentication attempts.

The Wazuh SIEM platform—deployed as the central monitoring solution—detected the attack in real time, generating alerts for failed authentication events, brute-force patterns, and ultimately the successful authentication. These alerts map to MITRE ATT&CK techniques T1110 (Brute Force) and T1078 (Valid Accounts).

This exercise demonstrates end-to-end detection capability: from simulated attack through alert generation, log correlation, and investigative analysis.

2. Lab Environment

Component	Role	OS	IP Address
Wazuh Manager	SIEM (Manager + Indexer + Dashboard)	Ubuntu Desktop 24.04	10.0.0.58
Ubuntu-Target	Monitored endpoint	Ubuntu Server 24.04	10.0.0.92
Attacker	SSH brute-force tool	Ubuntu Desktop (Wazuh VM)	10.0.0.58

Network: All systems bridged on the same subnet (10.0.0.0/24)

Note on Attacker Placement: Due to host hardware constraints (12 GB RAM, insufficient to run three concurrent virtual machines), the attack was executed from the Wazuh Manager VM itself rather than from a dedicated Kali Linux attacker VM. The detection logic, log generation, and alert behavior are functionally identical to those produced by an external attacker host. In a production or higher-resource lab environment, the attack would originate from a separate Kali Linux instance to more accurately model an external threat actor.

3. Attack Simulation

The brute-force attack was conducted using Hydra v9.5, an open-source credential-cracking tool.

Command executed:

```
bash
hydra -l vboxuser -P ~/mini-list.txt ssh://10.0.0.92
```

Wordlist contents:

```
text
password
123456
letmein
admin
qwerty
vboxuser
```

A larger follow-up run was also executed against the rockyou.txt wordlist to generate additional alert volume.

4. Result

Hydra successfully identified valid credentials for the user vboxuser:

```
text
[22][ssh] host: 10.0.0.92  login: vboxuser  password: [REDACTED]
1 of 1 target successfully completed, 1 valid password found
The attack completed in under 60 seconds, confirming weak password hygiene as a realistic and
rapidly exploitable risk.
```

5. Detection and Alerts

Wazuh generated the following alert types in response to the attack:

Rule ID	Description	Significance
5710	Attempt to login using a non-existent user	Reconnaissance / failed attempts
5712	Multiple authentication failures from the same source	Brute-force pattern
5760	sshd: authentication failed	Individual failed login
5763	sshd: brute-force attempt	Wazuh's built-in brute-force detection
5715	sshd: authentication success	Successful compromise
PAM (various)	PAM: user login failed	System-level auth rejection

6. The Critical Signal

The highest-priority alert is the sshd: authentication success that followed a sustained burst of failures. This sequence—many failures followed by a success—is the canonical indicator of a successful brute-force compromise and is escalated in real SOC workflows.

7. Alert Fields Captured

For the successful authentication alert, Wazuh recorded:

Field	Value
agent.name	ubuntu-target
agent.ip	10.0.0.92
rule.id	5715
rule.description	sshd: authentication success
rule.mitre.id	T1078
data.srcip	10.0.0.58
data.dstuser	vboxuser
full_log	Raw sshd entry showing successful authentication

8. MITRE ATT&CK Mapping

Technique ID	Technique Name	Tactic	Relevance
T1110	Brute Force	Credential Access	Repeated password attempts against SSH
T1110.001	Password Guessing	Credential Access	Specific sub-technique applicable
T1078	Valid Accounts	Initial Access / Persistence	Successful login using legitimate credentials
T1021.004	SSH	Lateral Movement	SSH as the access vector

9. Findings

9.1 What Worked

- Log forwarding: The Wazuh agent on the target forwarded SSH and PAM authentication events to the manager reliably.
- Detection rules: Out-of-the-box Wazuh rules correctly identified brute-force patterns and the subsequent successful login.
- MITRE mapping: Alerts were automatically mapped to ATT&CK techniques without custom configuration.
- Dashboard visibility: The attack was clearly visible in Security Events and the MITRE ATT&CK view.

9.2 Gaps Identified

- No automated response: Wazuh detected the attack but took no defensive action (no IP blocking, no account lockout trigger).
- No alerting integration: In a production environment, this would need to route to email, Slack, or a SOAR platform.
- Attacker location: Due to resource constraints, the attack originated from the manager host—this would be a Kali VM in a fully-resourced lab.

9.3 Lessons Learned

This exercise reinforced the value of SIEM visibility in detecting credential-based attacks. It also highlighted the gap between detection and response—identifying an attack is only half the battle; automated response, alerting, and remediation workflows are what actually reduce risk.

10. Recommendations

If this were a production system, the following remediations would be recommended:

1. Disable SSH password authentication — require SSH key pairs instead
2. Deploy fail2ban or a similar tool to auto-ban IPs after N failures
3. Enforce strong password policy — the compromised password would fail a basic complexity check
4. Enable MFA on SSH where feasible
5. Restrict SSH access — limit to trusted IPs via firewall rules
6. Configure Wazuh active response — auto-block source IPs when Rule 5712 or 5763 fires

7. Integrate alerts with an on-call notification channel

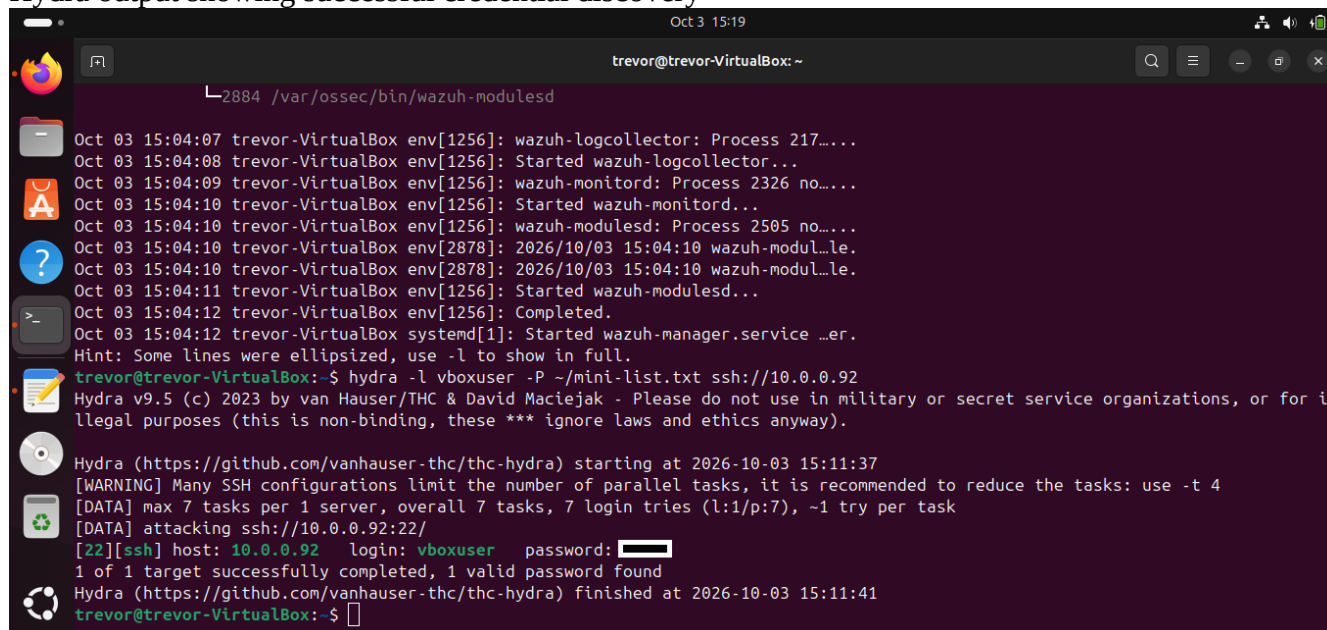
11. Conclusion

This exercise validated the Wazuh SIEM deployment's ability to collect authentication logs from a monitored Linux endpoint, detect a brute-force attack in progress, identify a successful credential compromise, map alerts to the MITRE ATT&CK framework, and present findings in a dashboard suitable for SOC analysis.

The project demonstrates a complete detection engineering workflow—from simulated attack through log ingestion, rule evaluation, alert triage, and incident documentation—mirroring the daily responsibilities of a SOC analyst.

Appendix A: Screenshots

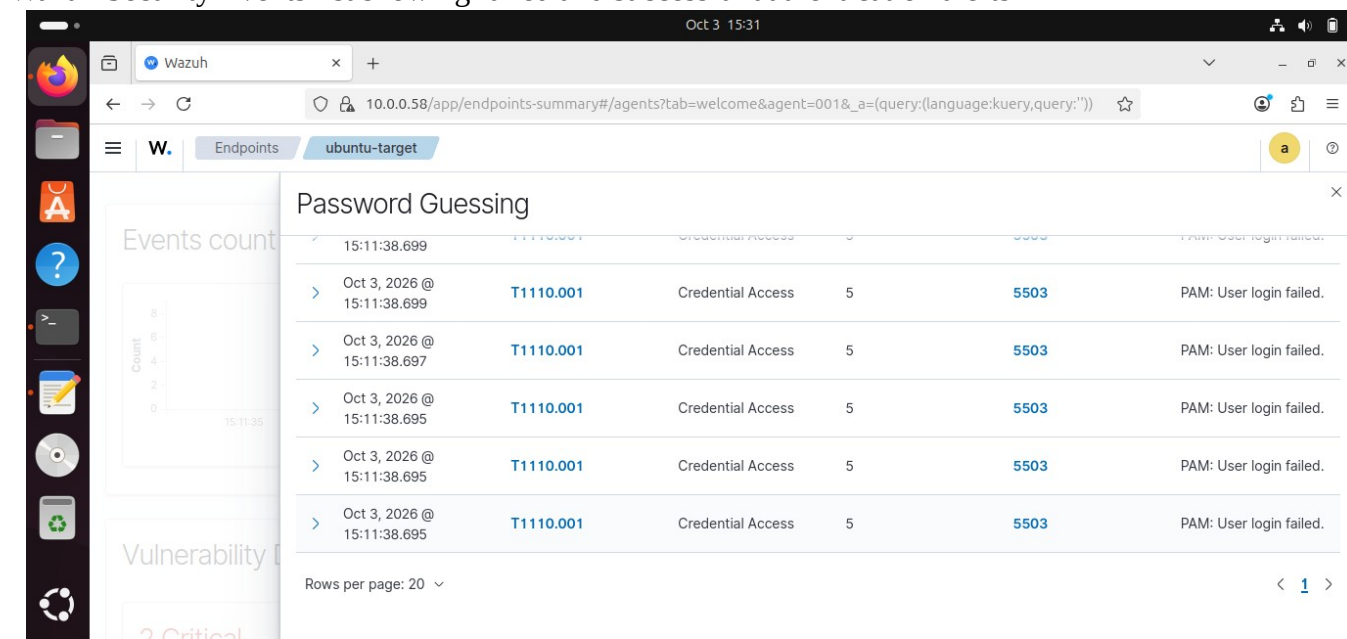
Hydra output showing successful credential discovery



```
Oct 3 15:19
trevor@trevor-VirtualBox: ~
└─2884 /var/ossec/bin/wazuh-modulesd
Oct 03 15:04:07 trevor-VirtualBox env[1256]: wazuh-logcollector: Process 217....
Oct 03 15:04:08 trevor-VirtualBox env[1256]: Started wazuh-logcollector...
Oct 03 15:04:09 trevor-VirtualBox env[1256]: wazuh-monitord: Process 2326 no....
Oct 03 15:04:10 trevor-VirtualBox env[1256]: Started wazuh-monitord...
Oct 03 15:04:10 trevor-VirtualBox env[1256]: wazuh-modulesd: Process 2505 no....
Oct 03 15:04:10 trevor-VirtualBox env[2878]: 2026/10/03 15:04:10 wazuh-modul...le.
Oct 03 15:04:10 trevor-VirtualBox env[2878]: 2026/10/03 15:04:10 wazuh-modul...le.
Oct 03 15:04:11 trevor-VirtualBox env[1256]: Started wazuh-modulesd...
Oct 03 15:04:12 trevor-VirtualBox env[1256]: Completed.
Oct 03 15:04:12 trevor-VirtualBox systemd[1]: Started wazuh-manager.service ..er.
Hint: Some lines were ellipsized, use -l to show in full.
trevor@trevor-VirtualBox: $ hydra -l vboxuser -P ~/mini-list.txt ssh://10.0.0.92
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-10-03 15:11:37
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 7 tasks per 1 server, overall 7 tasks, 7 login tries (l:1/p:7), ~1 try per task
[DATA] attacking ssh://10.0.0.92/
[22][ssh] host: 10.0.0.92 login: vboxuser password: [REDACTED]
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-10-03 15:11:41
trevor@trevor-VirtualBox: $
```

Wazuh Security Events list showing failed and successful authentication alerts



The screenshot shows the Wazuh Security Events dashboard. A modal window titled "Password Guessing" is open, displaying a list of authentication events. The events are categorized by "Credential Access" and show a "PAM: User login failed" message. The events are dated "Oct 3, 2026 @ 15:11:38.699" and "Oct 3, 2026 @ 15:11:38.697". The events are associated with the ID "T1110.001" and the IP address "5503".

Event ID	Timestamp	Category	Count	IP Address	Message
T1110.001	Oct 3, 2026 @ 15:11:38.699	Credential Access	5	5503	PAM: User login failed.
T1110.001	Oct 3, 2026 @ 15:11:38.697	Credential Access	5	5503	PAM: User login failed.
T1110.001	Oct 3, 2026 @ 15:11:38.695	Credential Access	5	5503	PAM: User login failed.
T1110.001	Oct 3, 2026 @ 15:11:38.695	Credential Access	5	5503	PAM: User login failed.
T1110.001	Oct 3, 2026 @ 15:11:38.695	Credential Access	5	5503	PAM: User login failed.

Alert detail (JSON/Table view) for the successful authentication

The screenshot displays the Wazuh Threat Hunting interface. The left pane shows a table of alerts for the agent 'ubuntu-target'. The right pane shows the 'Document Details' for a selected alert, displaying the JSON data.

Alerts Table:

timestamp	agent.name	rule.description
Oct 3, 2026 @ 15:11:40.663	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:40.661	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:40.657	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:40.655	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:40.653	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:40.649	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:40.637	ubuntu-target	PAM: Login sess...
Oct 3, 2026 @ 15:11:40.628	ubuntu-target	PAM: Login sess...
Oct 3, 2026 @ 15:11:40.628	ubuntu-target	sshd: authentic...
Oct 3, 2026 @ 15:11:38.699	ubuntu-target	PAM: User login ...
Oct 3, 2026 @ 15:11:38.699	ubuntu-target	PAM: User login ...

Document Details (JSON):

```
{  "_index": "wazuh-alerts-4.x-2026.10.03",  "_id": "IQSkA6EB72LHPDj-1DcW",  "_score": null,  "_source": {    "predecoder": {      "hostname": "Ubuntu-Target",      "program_name": "sshd",      "timestamp": "Oct 03 20:11:38"    },    "input": {      "type": "log"    },    "agent": {      "ip": "10.0.0.92",      "name": "ubuntu-target",      "id": "001"    }  },}
```

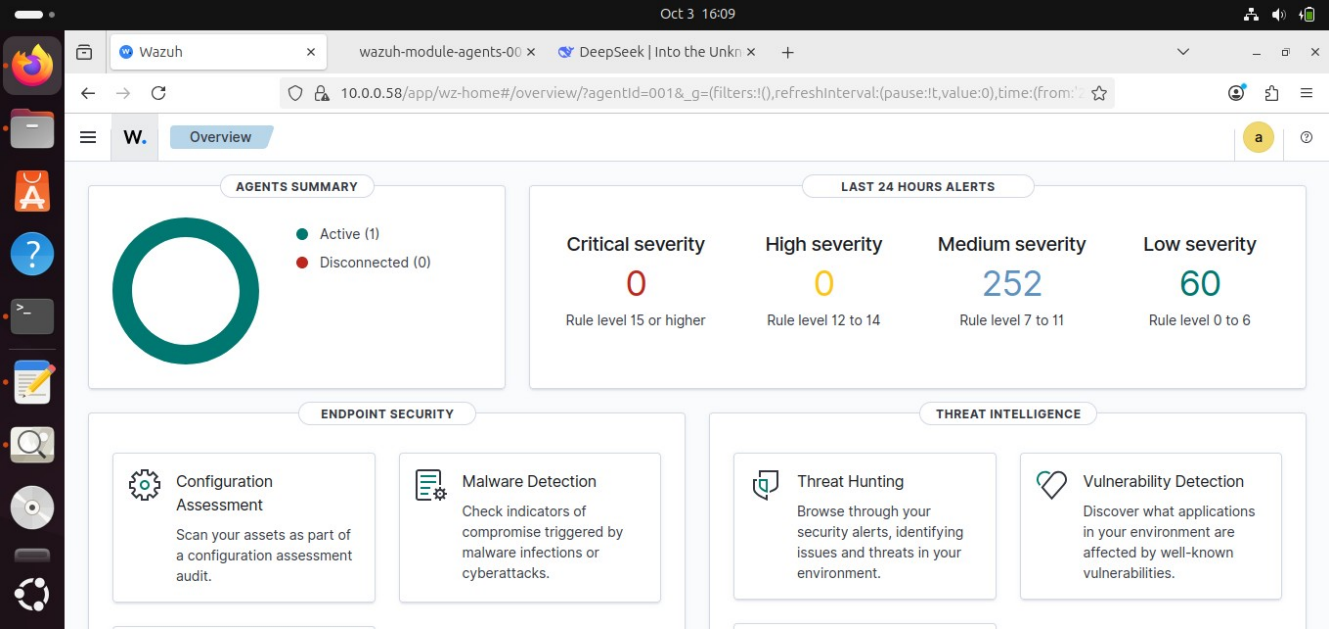
MITRE ATT&CK dashboard view showing T1110 / T1078

The screenshot displays the Wazuh MITRE ATT&CK dashboard. The table shows 14 hits for the agent 'ubuntu-target' across various MITRE ATT&CK techniques.

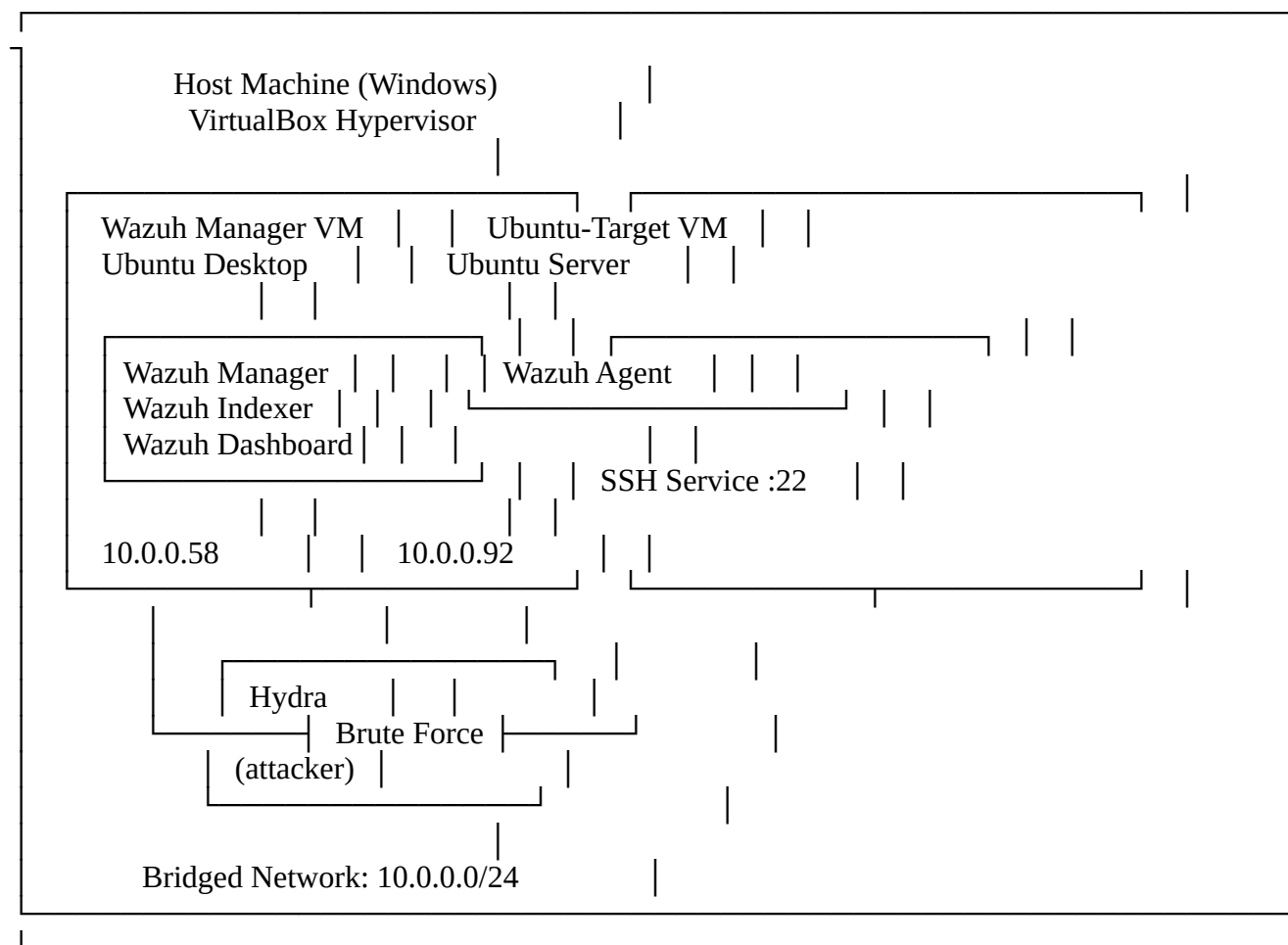
MITRE ATT&CK Hits Table:

timestamp	agent.name	rule.mitre.id	rule.mitre.tactic	rule.description	rule.level	rule.id
Oct 3, 2026 @ 15:11:40.663	ubuntu-target	T1110.001 T1078	Credential Access, Lateral Movement	sshd: authentic...	5	5760
Oct 3, 2026 @ 15:11:40.661	ubuntu-target	T1110.001 T1078	Credential Access, Lateral Movement	sshd: authentic...	5	5760
Oct 3, 2026 @ 15:11:40.657	ubuntu-target	T1110.001 T1078	Credential Access, Lateral Movement	sshd: authentic...	5	5760
Oct 3, 2026 @ 15:11:40.655	ubuntu-target	T1110.001 T1078	Credential Access, Lateral Movement	sshd: authentic...	5	5760
Oct 3, 2026 @ 15:11:40.653	ubuntu-target	T1110.001 T1078	Credential Access, Lateral Movement	sshd: authentic...	5	5760
Oct 3, 2026 @ 15:11:40.649	ubuntu-target	T1110.001 T1078	Credential Access, Lateral Movement	sshd: authentic...	5	5760
Oct 3, 2026 @ 15:11:40.628	ubuntu-target	T1078	Defense Evasion, Persistence, Privileg...	PAM: Login sess...	3	5501
Oct 3, 2026 @ 15:11:40.628	ubuntu-target	T1078 T1021	Defense Evasion, Persistence, Privileg...	sshd: authentic...	3	5715
Oct 3, 2026 @ 15:11:38.699	ubuntu-target	T1110.001	Credential Access	PAM: User login ...	5	5503
Oct 3, 2026 @ 15:11:38.699	ubuntu-target	T1110.001	Credential Access	PAM: User login ...	5	5503
Oct 3, 2026 @ 15:11:38.697	ubuntu-target	T1110.001	Credential Access	PAM: User login ...	5	5503

Agents view confirming ubuntu-target is Active



Lab architecture diagram



Appendix B: Commands Used

Attack

```
hydra -l vboxuser -P ~/mini-list.txt ssh://10.0.0.92
```

Agent installation (on target)

```
sudo WAZUH_MANAGER='10.0.0.58' WAZUH_AGENT_NAME='ubuntu-target' dpkg -i ./wazuh-agent_4.12.0-1_amd64.deb
```

Agent service management

```
sudo systemctl daemon-reload
```

```
sudo systemctl enable wazuh-agent
```

```
sudo systemctl start wazuh-agent
```